

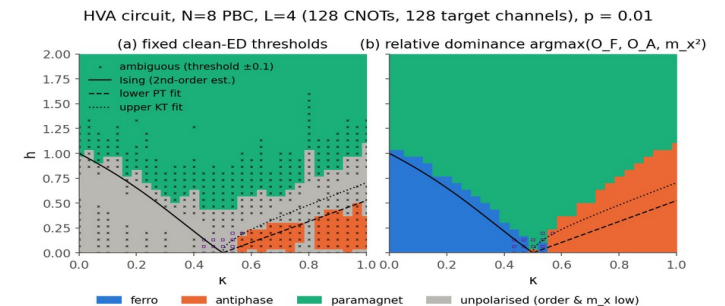
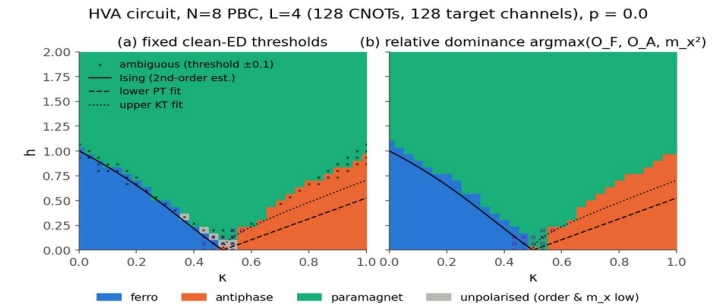
The ANNNI phase diagram under CNOT-target depolarizing noise

Exact diagonalisation, a 128-CNOT variational circuit, and what 1% and 5% noise do to the phases

Benjamin Kaganovitch · Dimitrii Kashtanov · Yan Ivanov · Frederick Pu

Q-SITE Hacks 2026 · Quantum Coalition Scientific Open Challenge

github.com/BenjaminKaganovitch/qsites-2026-open-challenge



Model, order parameters and reference curves

$$H = -\sum_i Z_i Z_{i+1} + \kappa \sum_i Z_i Z_{i+2} - h \sum_i X_i$$

Ferromagnet

κ small, h small. All Z aligned;
 $S(0) = N$.

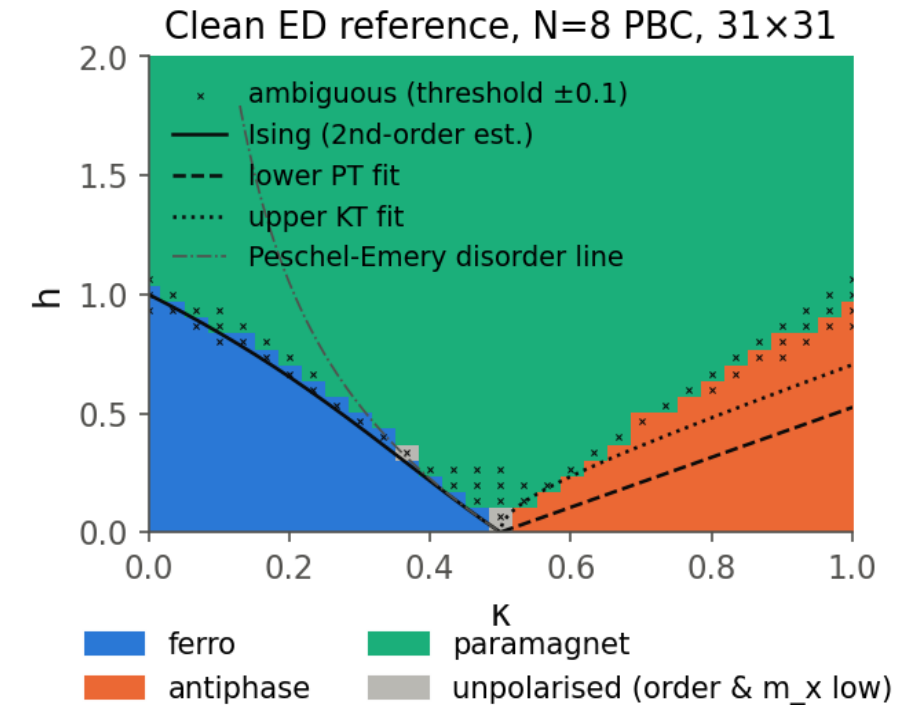
Antiphase

$\kappa > \frac{1}{2}$, h small. Period-4
pattern $\uparrow \uparrow \downarrow \downarrow$; $S(\pi/2) = N/2$.

Paramagnet

h large. Spins follow the field;
 $m_x \rightarrow 1$.

- Finite rings keep the spin-flip symmetry, so $\langle Z \rangle = 0$ even in ordered phases: we classify with correlations.
- Structure factor $S(q) = (1/N) \sum e^{iq(i-j)} \langle Z_i Z_j \rangle$; floor-subtracted $O_F = (S(0)-1)/(N-1)$, $O_A = (S(\pi/2)-1)/(N/2-1)$, plus m_x .
- Overlay curves are approximate thermodynamic references: Ising estimate, Pokrovsky-Talapov line $1.05(\kappa - \frac{1}{2})$, KT line $1.05\sqrt{((\kappa - \frac{1}{2})(\kappa - 0.1))}$.
- Thresholds $T = 0.541$, $T_x = 0.490$ are calibrated on limiting-case anchor regions, not on the curves.



Exact ground states, $N = 8$ periodic, 31×31 grid. Ferro (blue), antiphase (orange), paramagnet (green).

Methods and validation

1

Exact reference

Sparse Lanczos (six lowest eigenpairs), $N = 8, 10, 12$ on 31×31 grids plus 21×401 fine cuts. Warm and cold starts agree to $1e-13$.

2

Variational circuit

Hamiltonian-variational ansatz: $|+\rangle^N$ then $L = 4$ layers of (ZZ_{nn}, ZZ_{nnn}, X) rotations, 12 parameters, symmetry preserving. Each ZZ bond is CNOT-RZ-CNOT: 128 CNOTs.

3

Noise model

DepolarizingChannel(p) on the target after every CNOT on default.mixed; 128 channels verified on the tape. Same parameters, grid and thresholds at $p = 0, 0.01, 0.05$.

98.9%

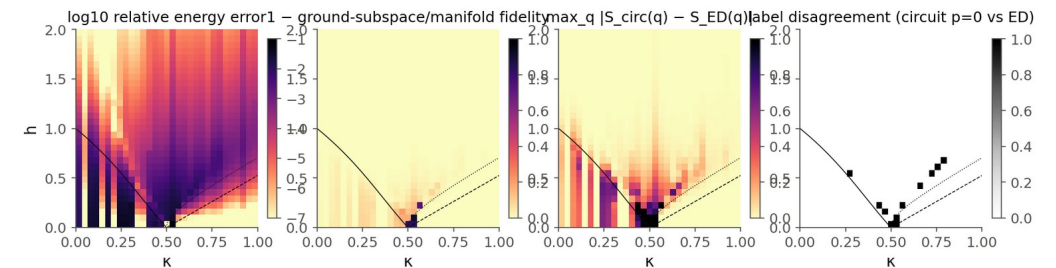
circuit labels agree
with exact
diagonalisation

4e-5

median relative
energy error at $p = 0$

1e-14

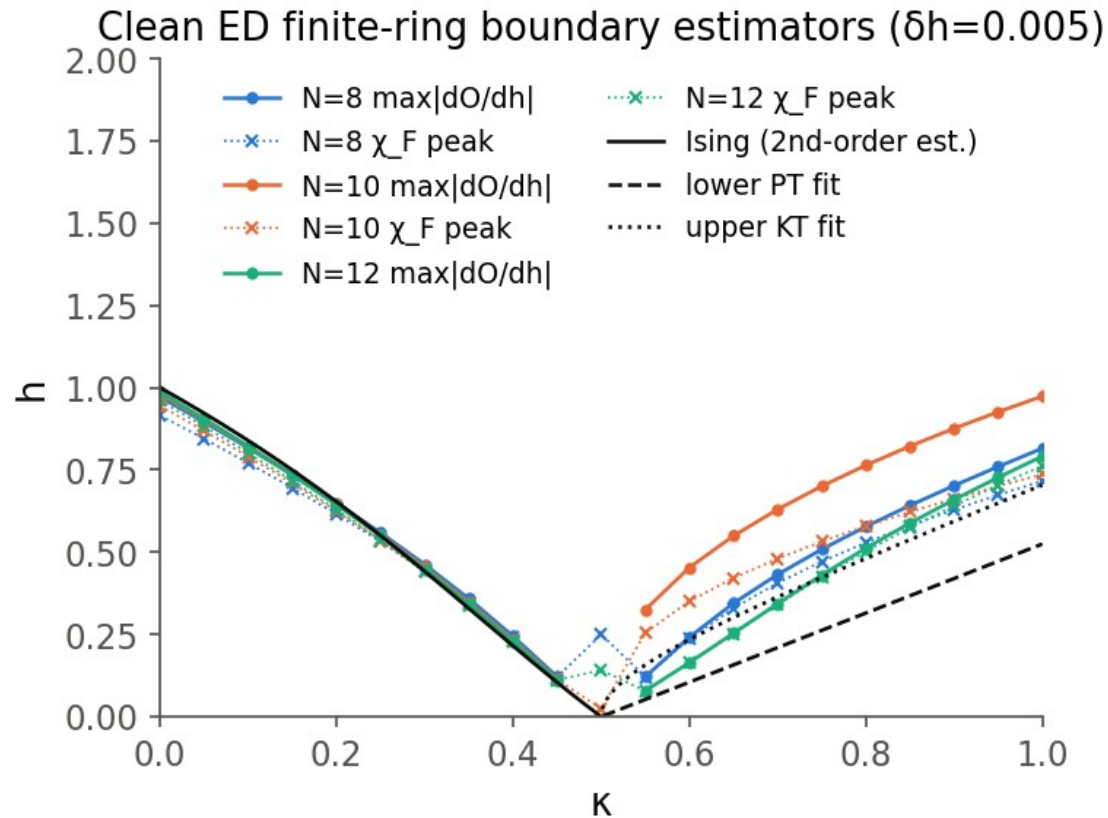
engines match
PennyLane
default.qubit /
mixed



Clean circuit vs exact ground states: relative energy error, 1 - fidelity, ambiguity and label agreement.

- A hardware-efficient RY/CNOT ring stalled at 1-3% energy error near the boundaries even at depth 6; the HVA does not.
- Parameters optimised at $p = 0$ by L-BFGS with exact adjoint gradients, warm-started sweeps and restarts.

Exact phase diagram and finite-size behaviour



Boundary estimators from exact diagonalisation at $N = 8, 10, 12$ with the Ising, PT and KT reference curves.

0.616 $\rightarrow$ 0.629

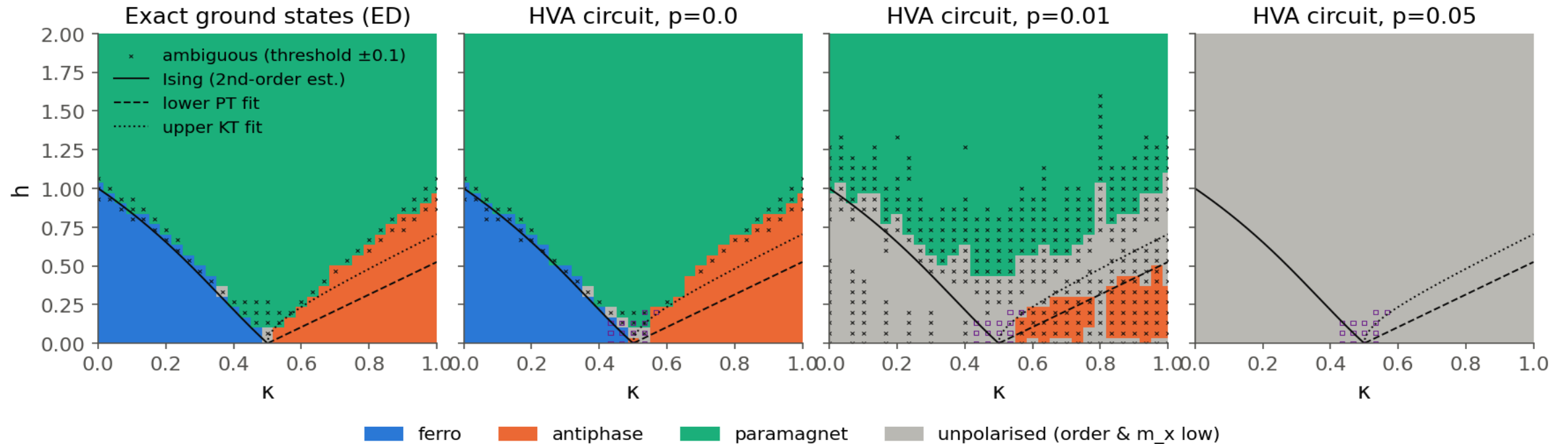
ferro boundary at $\kappa = 0.2$, $N = 8 \rightarrow 12$
(DMRG 0.639)

0.562 $\rightarrow$ 0.432

antiphase boundary at $\kappa = 0.75$, $N = 8$
 $\rightarrow 12$ (KT fit 0.423)

- Ferro-para boundary is almost size independent and tracks the Ising estimate; the fidelity-susceptibility peak moves toward the DMRG value.
- Antiphase boundary drifts down strongly with N toward the KT line, the expected slow finite-size behaviour of a KT-like transition.
- $N = 10$ cannot hold a period-4 pattern and is anomalous (a commensurability effect, not thermodynamic physics).
- Floating phase: at $N = 12$ the entanglement entropy peaks near the KT line, but the ordering wavevector stays locked at $\pi/2$. We call it unresolved rather than claim it.
- A label-free k-means on $(S(q)/N, m_x)$ reproduces the rule labels at 99.1%.

Effect of depolarising noise at $p = 0.01$ and 0.05



Same circuit, parameters, grid and thresholds. Left to right: exact reference, circuit at $p = 0, 0.01, 0.05$. Purple: clean fidelity < 0.9 ; x: threshold-ambiguous.

0.60

median contraction of order parameters at $p = 0.01$

0%

of ferro points keep their label at $p = 0.01$ (antiphase 46%, para 89%)

0.005

median shift of derivative-peak boundaries at $p = 0.01$

0.0047

purity at $p = 0.05$, vs 0.0039 for the maximally mixed state

Channel-insertion analysis of noise sensitivity

First-order response of each order parameter, summed over all 128 channel locations: $d \ln O / dp$

-99

ferromagnet, O_F

-69

antiphase, O_A

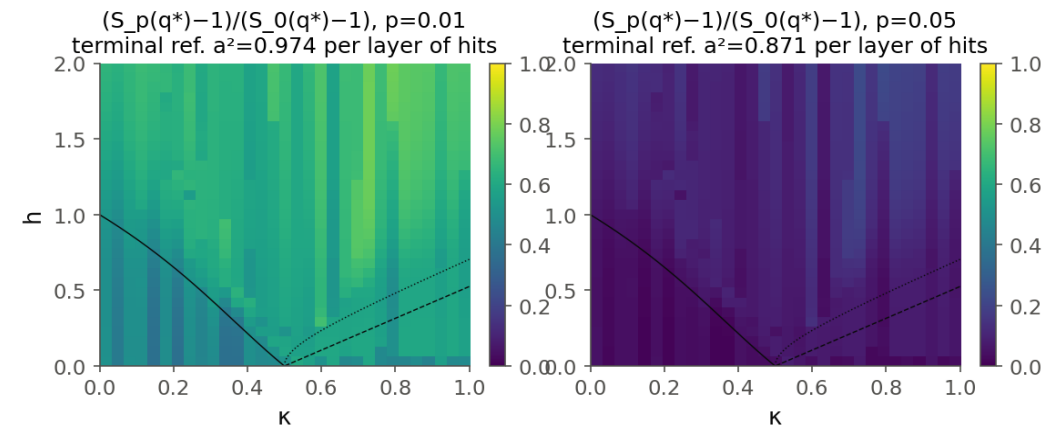
-48

paramagnet, m_x

-43

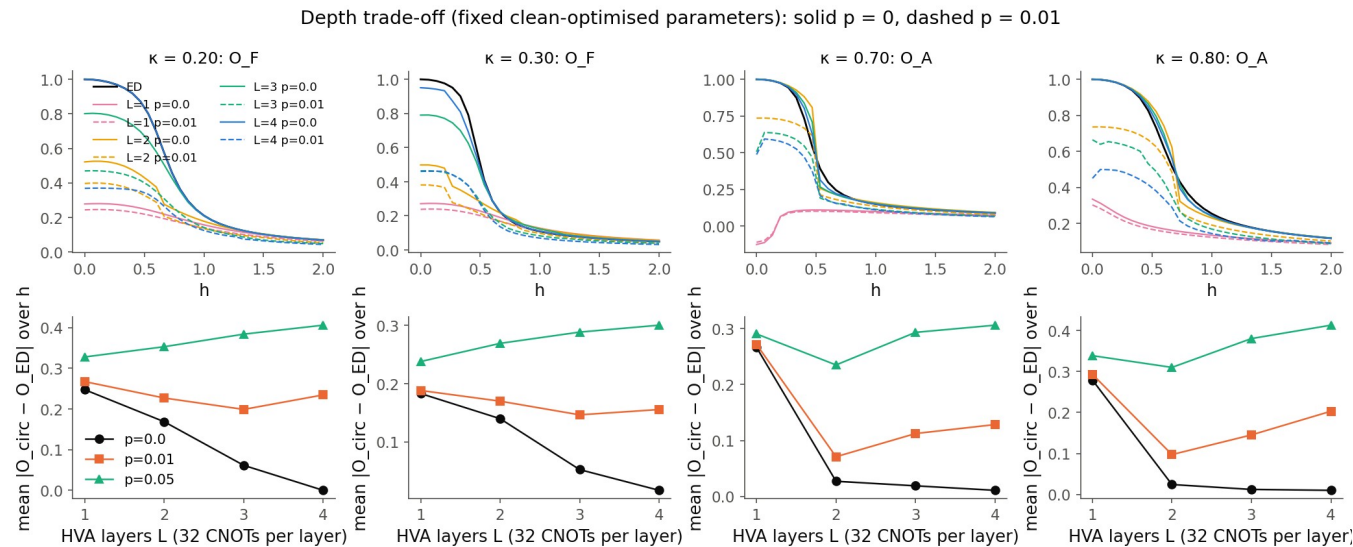
if every hit acted as end-of-circuit noise

- Mid-circuit errors are carried through later CNOT-RZ-CNOT gadgets, so they damage ZZ order about $2.3\times$ more than terminal noise would.
- Early layers contribute most: for the ferro point, layer 1 gives -27 and layer 4 gives -17.
- The GHZ-like ferro ring loses all $N(N-1)$ positive correlators to a single propagated X/Y error; O_A mixes signs, so damage partly cancels.
- First-order theory gets the sign of the boundary shift right but underestimates it 1.5-4 $\times$: $p = 0.01$ is already nonlinear at this depth.
- This ranking is specific to this circuit and diagnostic; it reverses the organiser's hypothesis but is not a law about the phases.



Ratio of noisy to clean structure factor at the dominant wavevector: $p = 0.01$ (left) and $p = 0.05$ (right).

Circuit depth, re-optimisation and clustering



Clean error and noisy error against depth $L = 1-4$ on four κ cuts. The best depth falls as p grows.

- Shallow circuits are wrong even without noise; deep ones collect more noise. The antiphase is accurate at $L = 2$ (error 0.024 at $\kappa = 0.8$), the ferromagnet needs $L = 4$ (error 0.17 at $L = 2$).
- At $p = 0.01$ the best depth is $L = 3$ for ferro and $L = 2$ for antiphase: a second reason the ferromagnet suffers most.

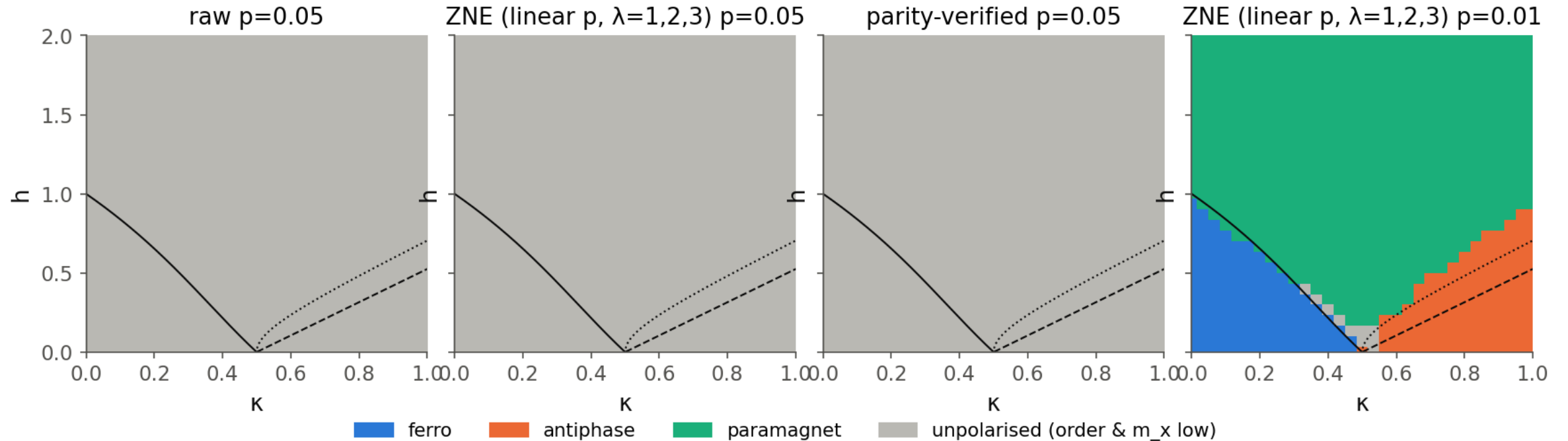
Noise-aware re-optimisation

Minimising $\text{Tr}(\text{Hp}_p)$ with exact gradients through the channels lowers the noisy energy by 0.2–1.0 per cut, but the order parameters move by only -0.016 to $+0.028$. The optimiser gains energy through m_x by leaving the ground state (clean energy -5.83 vs -6.40 at $\kappa = 0.2$, $h = 0$). Energy-optimal noisy states are not diagnostic-optimal.

Label-free clustering

k-means on $(S(q)/N, m_x)$ reproduces the clean rule labels at 99.1%: a consistency check, not independent validation.

Error mitigation by extrapolation and parity checks



Fixed thresholds. Raw $p = 0.05$; ZNE at $p = 0.05$ (fails); parity-verified $p = 0.05$; ZNE at $p = 0.01$ (Richardson, $\lambda = 1, 2, 3$, $p = 0$ truth held out).

0.063 $\rightarrow$ 0.011

median correlator error at $p = 0.01$
after ZNE

70% $\rightarrow$ 98%

label agreement with the clean circuit
after ZNE

$\times 19$

variance amplification $\Sigma \gamma^2$; at $p = 0.05$
ZNE fails (signal $\approx 1\%$)

71%

parity-verification acceptance at $p = 0.01$;
error only 0.063 $\rightarrow$ 0.053

Conclusions and limitations

1

Accurate clean diagram: on 8–12-site rings; the ferro side converges quickly, the antiphase side drifts toward the KT line.

2

1% noise removes the ferromagnet: under fixed thresholds, while derivative-defined boundaries stay put; 5% leaves no usable signal.

3

Depth vs noise is the real trade-off: the state that needs the full circuit carries the most channels and suffers first.

4

ZNE turns $p = 0.01$ back into a correct diagram: parity checks help little; noise-aware re-optimisation buys energy, not diagnostics.

Quench dynamics: a Trotterised quench from $|+\rangle$ shows a sharp Loschmidt-rate peak only deep in the ordered side ($h \leq 0.4$); at $p = 0.01$ the channel error is about $8\times$ the Trotter error, so the dynamical signature does not survive noise.

Limitations

- Small N (≤ 12); overlays are approximate
- $\kappa \approx 0.5$ masked for preparation quality
- Fixed-parameter protocol on the full grid; re-optimisation only on cuts
- Exact simulation, no shot noise

github.com/BenjaminKaganovitch/qsite-2026-open-challenge

